

Lecture 11

11/09/24 10:16 AM

- $E[g(X,Y)]$

$$E[aX+bY+c] = a \cdot E[X] + b \cdot E[Y] + c$$

have to compute marginals

$$= \sum_{x,y} (ax+by+c) \cdot p_{xy}(x,y)$$

$$= a \cdot \sum_{xy} x \cdot p_{xy}(x,y) + b \cdot \sum_{xy} y \cdot p_{xy}(x,y) + c$$

$$= a \cdot E[X] + b \cdot E[Y] + c$$

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) \cdot p_{xy}(x,y)$$

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- Finding $p_Z(\cdot)$ where $Z = g(X,Y)$:

Suppose $Z = g(X)$.

$$\Rightarrow p_Z(z) = \sum_{\{x: g(x)=z\}} p_X(x)$$

Now suppose $Z = g(X,Y)$

$$\Rightarrow p_Z(z) = \sum_{\{x,y: g(x,y)=z\}} p_{xy}(x,y)$$

$$\therefore \text{We can then calculate } E[Z] = E[g(X,Y)] = \sum_Z z \cdot p_Z(z)$$

$$= \sum_Z \sum_{\{x,y: g(x,y)=z\}} z \cdot p_{xy}(x,y)$$

since every xy corresponds to a z.

$$= \sum_{x,y} g(x,y) \cdot p_{xy}(x,y)$$

=

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) \cdot f_{xy}(x,y) \cdot dx dy$$

Q. Suppose X and Y are continuous independent random variables.

Let $Z_1 = \max(X,Y)$ and $Z_2 = \min(X,Y)$

ans: $Z = \max(X,Y)$

$$F_Z(z) = P(Z \leq z) = P(\max(X,Y) \leq z) = P(X \text{ and } Y \leq z)$$

$$= F_X(z) \cdot F_Y(z)$$

$Z = \min(X,Y)$

HW: X and Y are exponential with parameters λ_1 and λ_2 .

$Z = \min(X,Y)$

$\Rightarrow Z$ is also exponential with parameter $\lambda_1 + \lambda_2$.

- COVARIANCE OF X AND Y .

$$\text{cov}(X,Y) = E[(X-E[X])(Y-E[Y])] = E[XY] - E[X]E[Y]$$

Covariance = 0 $\Rightarrow$ uncorrelated

($\equiv$ independent?)

- CONDITIONING WITH RANDOM VARIABLES:

Consider $\{\omega \in \Omega : X(\omega) = x\} = \{X = x\}$

$$P(X=x|A) = \frac{P(\{X=x\} \cap A)}{P(A)}$$

$$P_{X|A}(x) = \frac{P(\{X=x\} \cap A)}{P(A)}$$